

Pattern Detectives

GRADES
3-5

How machines notice similarities.



01 / NOTICE

Find useful features.

02 / SORT

Explain your rule.

03 / CHECK

Keep people in charge.

PUBLIC STUDENT LESSON / 45-60 MINUTES

No AI account required | Speak, point, draw, write, or use a scribe

PATTERN DETECTIVES / YOUR MISSION

Pattern Detectives



Notice similarities. Explain your choices. Check a machine's prediction.

Your detective mission

People sort things using rules. Some AI systems learn patterns from examples and use them to predict a group for a new item. A prediction still needs checking.

Words we will use

Pattern	A similarity, repeating order, or relationship we can notice.
Feature	Something about an item: its shape, material, size, or use.
Category	A group of items that fit a rule.
Classify	Sort items into groups.
Example / label	An item / the name or group someone gives it.
Human judgment	A person thinks carefully, checks, and decides.

I can...

Notice features; explain a sorting rule; compare another rule; describe why examples matter; give a reason people should check a result.

PATTERN DETECTIVES / NOTICE + SORT

Pattern Detectives



Choose a rule. Sort the same eight objects into Group A or Group B.

Choose a rule you can explain

You may use what an object is made of, what it does, or another clear feature. Do not use color alone. You can write, point, or sort real objects.

Backpack	Group: A / B	Feature / reason: _____
Basketball	Group: A / B	Feature / reason: _____
Book	Group: A / B	Feature / reason: _____
Bicycle	Group: A / B	Feature / reason: _____
Tablet	Group: A / B	Feature / reason: _____
Tennis shoe	Group: A / B	Feature / reason: _____
Water bottle	Group: A / B	Feature / reason: _____
Paintbrush	Group: A / B	Feature / reason: _____

My sorting rule is...

The hardest object to sort was... because...

Activity 1 | More than one rule may work; explain yours

PATTERN DETECTIVES / COMPARE + CREATE

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Try another rule with the same objects. Ask a partner to test it.

My first rule:

My partner's rule (or my second rule):

What is similar? What is different?

Could both rules work? Explain using one object.

Improve one rule so another person can follow it.

Test your rule

Ask a partner to sort one new object. If the rule is unclear, change it and test again. Explain why your change helps.

PATTERN DETECTIVES / EXAMPLES + LABELS

Pattern Detectives



Imagine choosing examples for a group called "Useful for learning."

A model, not a real AI tool

Choose labels for the objects below. Explain the learning activity you have in mind. Different choices may make sense in different situations.

Book	YES / NO	Pencil	YES / NO
Video game controller	YES / NO	Magnifying glass	YES / NO
Notebook	YES / NO	Headphones	YES / NO
Soccer ball	YES / NO	Map	YES / NO

Why did you choose those labels? Could a partner choose differently?

Missing examples can mislead

Imagine the only examples labeled "useful for learning" are books, notebooks, and maps. A pretend model notices paper and guesses headphones are not useful. That guess misses learning by listening.

What example or information is missing? What should a person check?

PATTERN DETECTIVES / HUMAN JUDGMENT CHECK

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An important grouping or recommendation needs a person to review it.

1 / An unfair guess

A pretend machine labels every student wearing glasses as "likes reading." Glasses do not tell us a person's interests.

What is missing or wrong? What should a person check or change?

2 / A difficult image

A photo system puts an object in the wrong group because the photo is dark.

What is missing or wrong? What should a person check or change?

3 / A narrow recommendation

A library app suggests only books like the first book a student borrowed.

What is missing or wrong? What should a person check or change?

PATTERN DETECTIVES / REFLECT + TAKE RESPONSIBILITY

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Explain what you learned. A person stays in charge of important decisions.

One thing I learned about patterns is...

Examples and labels can change a machine's result because...

A person should check the result because...

A question I still have is...

My responsible AI promise

☐ I will protect private information.

☐ I will check important information.

☐ I will question mistakes or unfair groupings.

☐ I will use technology to support my own thinking.

☐ I will ask a trusted adult when I am unsure.

Exit ticket

A machine can notice patterns by _____.

A person should check because _____.

Reflect + extend | Explain aloud, draw, or write; reasoning matters